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 - ▶ e.g., (a) thermal variations on surface of a planet depend on dynamics of planetary system
 - ▶ (b) properties of a molecule at a point of time depend on relative location of nuclei and electrons

scales in Nature : **Length** scales

scales in Nature : time scales

scales in Nature : **energy** scales

scientific modeling :

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For **most** objects that we encounter,

Newtonian or classical mechanics is a perfect scientific theory

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$\therefore$ EM radiation consists of sinusoidal fluctuating electric and magnetic fields

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 $\forall t \in \mathbb{R}, \vec{r} \in \mathbb{R}^3$
 - Invariance of laws of motion w.r.t. such a transformation implies
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3. rotation of the frame: $g_3(t, \vec{r}) = (t, G_R(\vec{r})); \quad \forall t \in \mathbb{R}, \vec{r} \in \mathbb{R}^3$
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$$\implies \dot{\vec{r}} = \vec{v}_0; \quad \vec{r}(t) = \vec{v}_0 t + \vec{r}_0$$

where $\vec{v}_0$ and $\vec{r}_0$ are initial velocity and initial position (at $t = 0$)

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future is completely determined by the current state of the system and is in principle calculable, given enough computing power

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2. object falling under gravity far from the surface of the Earth; $m\ddot{\vec{r}} = -m\vec{g}\frac{r_0^2}{r^2}$;
where r_0 is the radius of the Earth

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totality of all possible phase paths constitutes a phase portrait or phase diagram

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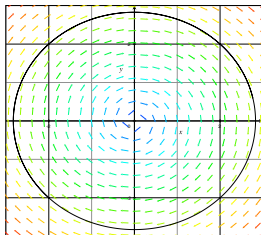
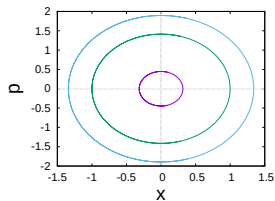
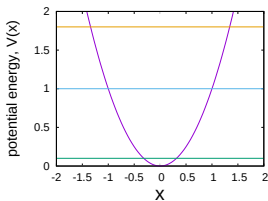
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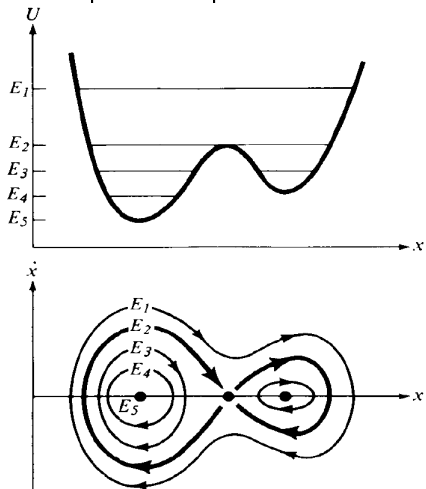
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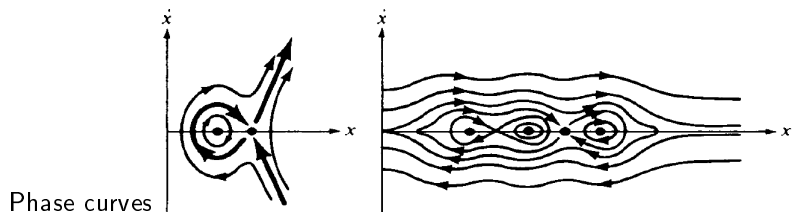
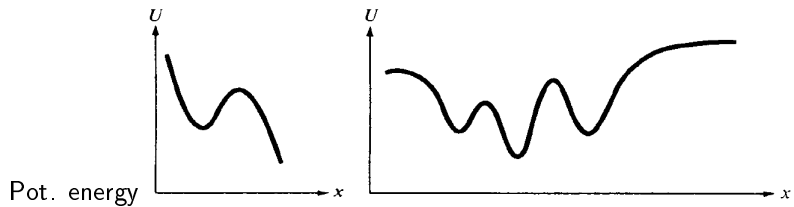
$$\text{or, } \ddot{x} + \omega_o^2 x = 0 \quad (x_0 = 0; \omega_o^2 = \frac{k}{m})$$

phase plane portrait for : $\dot{x} = y$; $\dot{y} = -x$
 $y \equiv$ momentum p and constants (mass and force constant, etc.) are unity;

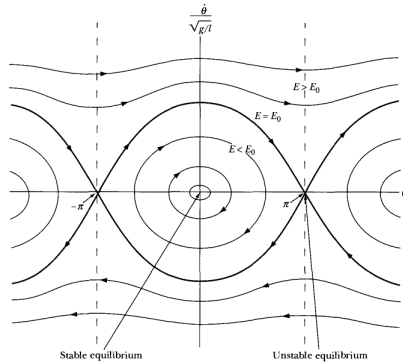
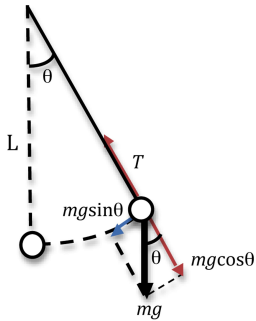


Phase space description :

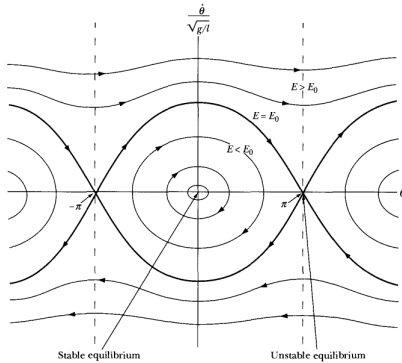
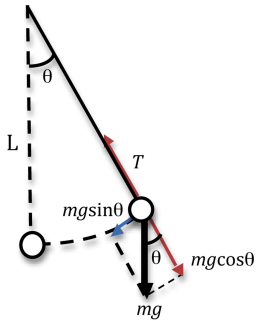




frictionless simple pendulum, $\ddot{\theta} = -\frac{g}{l} \sin \theta$, and its dynamics

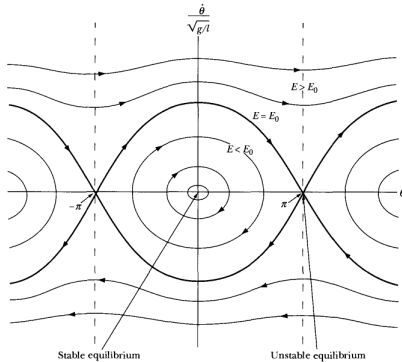
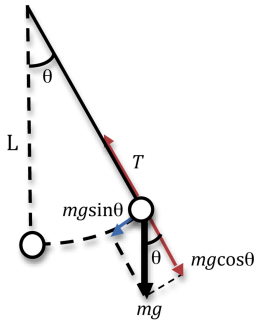


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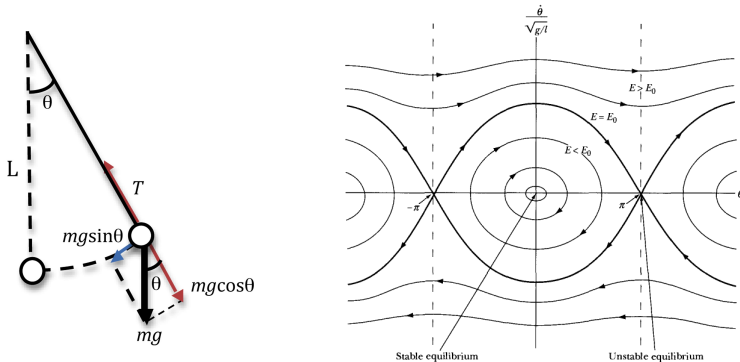
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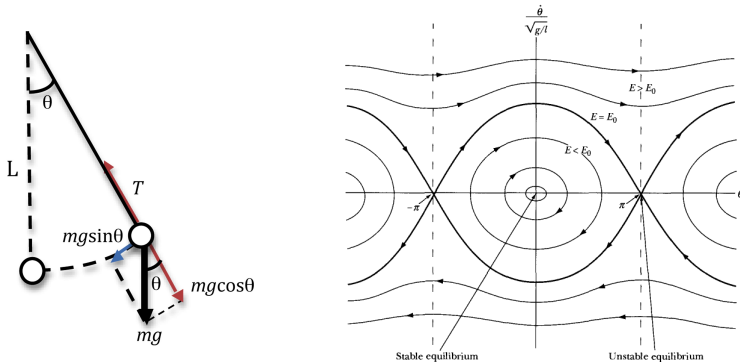
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integrating, $\frac{1}{2} \dot{\theta}^2 - \omega^2 \cos \theta = C$

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For example, in phase space,

Hamilton's equations of motion: $\dot{q}_i = \frac{\partial H}{\partial p_i}$; $\dot{p}_i = -\frac{\partial H}{\partial q_i}$, $\forall i = 1, 2, \dots, 3N$

(H = total energy)

are invariant in form as coordinates are transformed from one to another description

need for a generalised system of coordinates and equations of motion invariant w.r.t. change of coordinates

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Generalised coordinates: To work towards invariant equations of motion it is useful to talk in terms of generalised position and momentum coordinates q_i and p_i respectively, for $i = 1, 2, \dots, 3N$

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The answer comes from Hamilton's equations

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NOTE: Since we have partial derivatives of $\mathcal{L}(q, \dot{q}, t)$, we treat q , $\dot{q}$ and t as independent variables

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Evaluating a property, $\mathcal{P}(\vec{p}, \vec{q})$: $\langle \mathcal{P} \rangle = \int \mathcal{P}(\vec{p}, \vec{q}) f(\vec{p}, \vec{q}) d\vec{p} d\vec{q}$

where $f(\vec{p}, \vec{q})$ is the phase space distribution function : $\int f(\vec{p}, \vec{q}) d\vec{p} d\vec{q} = 1$

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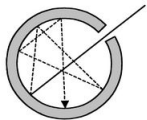
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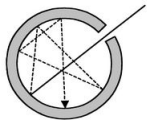
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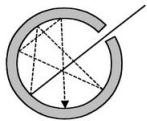




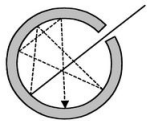
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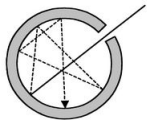
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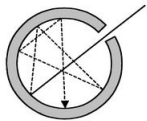
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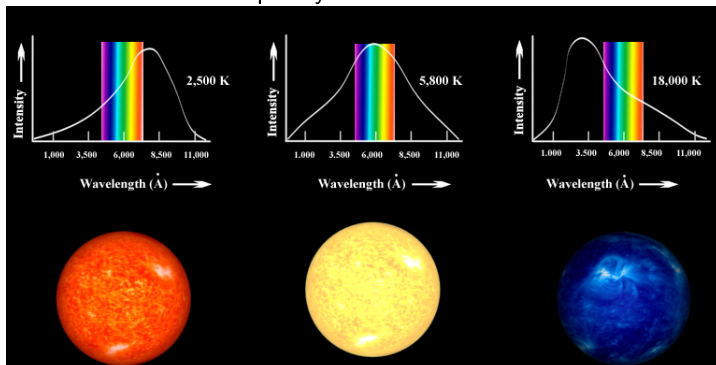


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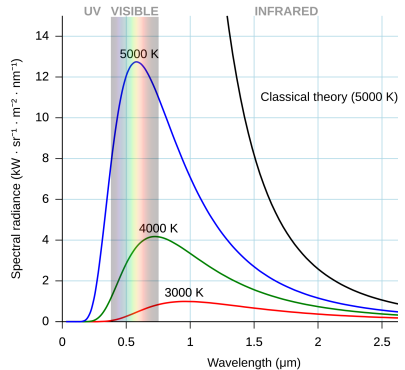
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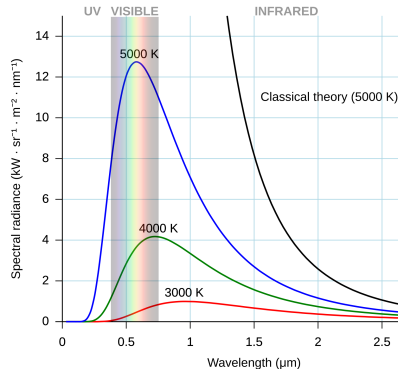
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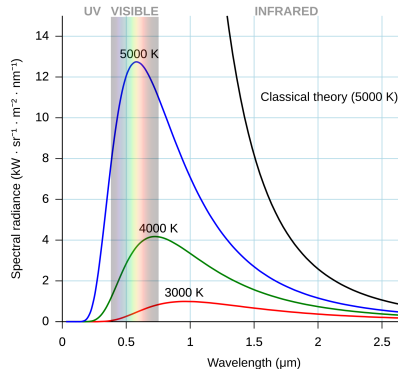
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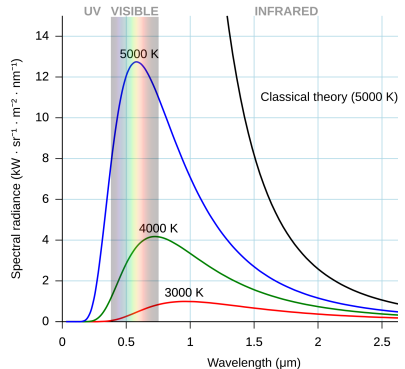
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- ▶ explanation of new formula required introduction of discrete packets or quanta of energy exchanged between matter and radiation, $E = h\nu$

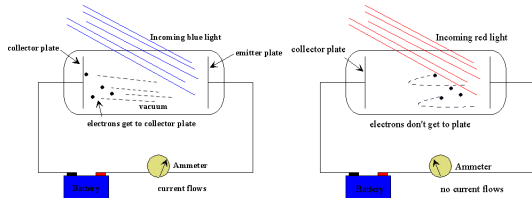
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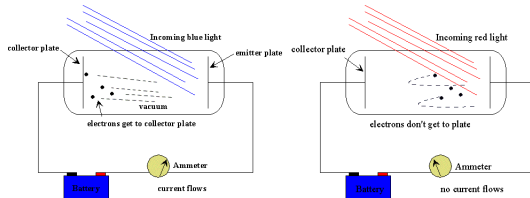
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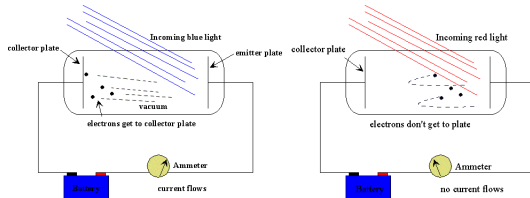
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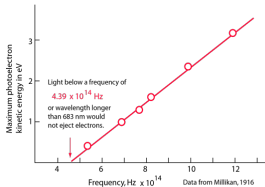
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electrons ejected by the blue light are getting to the collector plate, evidently the potential supplied by the battery is less than V_{stop} for blue light
(data shown for Na)

Einstein $\implies$ kinetic energy of emitted electron depends on the energy of the photon that hit the metallic surface and an effective "ionization energy" of the metal, $KE = h\nu - \Phi$

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Threshold frequency corresponds to a particle-light (photon) with the lowest energy needed to satisfy this work function

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where $\Phi = h\nu_0$, work function of the metal (energy required to free an electron) and ν_0 = threshold frequency

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To increase number of electrons ejected, it is necessary to increase the number of photons (intensity)

Atomic Spectra

A continuous spectrum is produced by an incandescent solid or gas at high pressure

Atomic Spectra

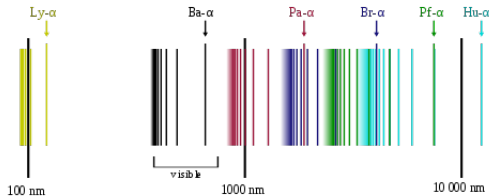
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Gases heated to incandescence were found by Bunsen, Kirchhoff and others to emit light with a series of sharp wavelengths. The emitted light analyzed by a spectrometer appears as a multitude of narrow bands of color. These so called line spectra are characteristic of the atomic composition of the gas.



spectral series of hydrogen, on a logarithmic scale

The series can be phenomenologically described via the relationship :

$$\bar{\nu} = R \left(\frac{1}{n_1^2 - n_2^2} \right)$$

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Paschen series : $n_1 = 3$; n_2 varies from 4 to ∞

Energy of a photon, $E = h\nu$

wave-particle duality : Light demonstrates both wavelike and particle-like behavior

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All forms of electromagnetic radiation share these properties

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The emitted light can be either continuous or discrete (from specific types of excited atoms).

wave nature of matter :

wave nature of matter : Davison-Germer experiment -
diffraction of low energy electrons from Ni crystal

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De Broglie Hypothesis : Wave-particle dual nature of matter

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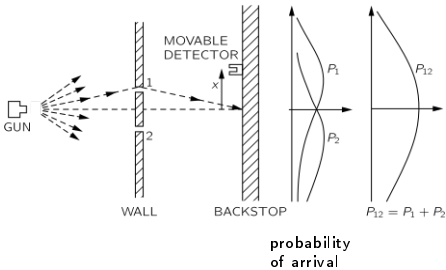
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$$p = \frac{h}{\lambda}$$

Bullets versus waves : Double slit

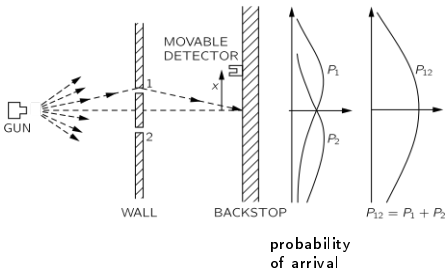
Bullets versus waves : Double slit

Experiment with bullets

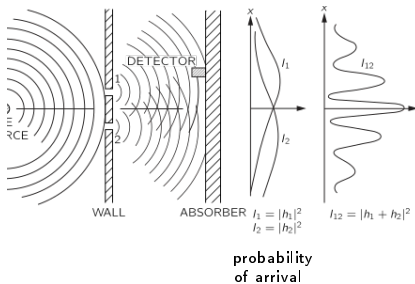


Bullets versus waves : Double slit

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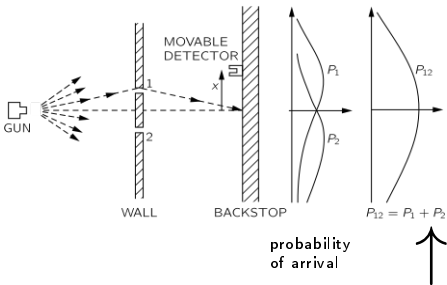


with waves (Young, 1801)

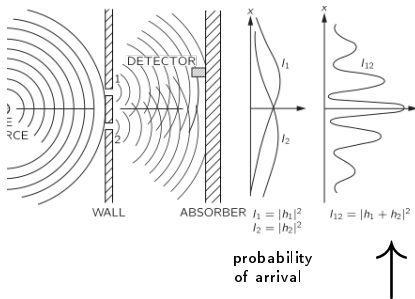


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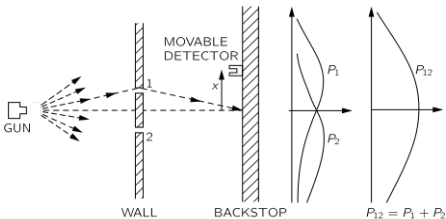


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Bullets versus waves : Double slit

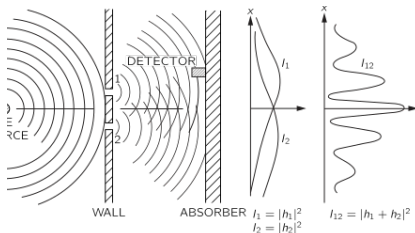
Experiment with bullets



probability
of arrival

↑
probabilities
add together
no interference

with waves (Young, 1801)

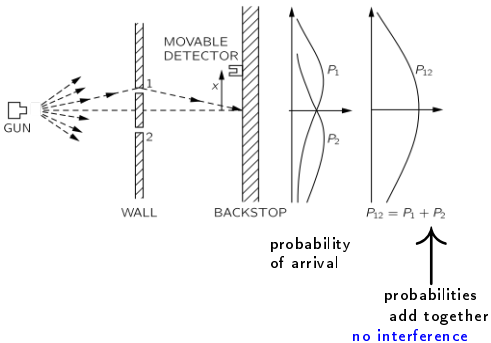


probability
of arrival

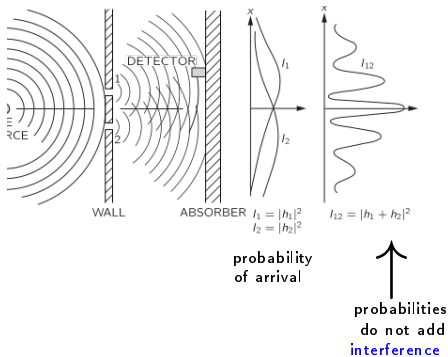
↑
probabilities
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Bullets versus waves : Double slit

Experiment with bullets

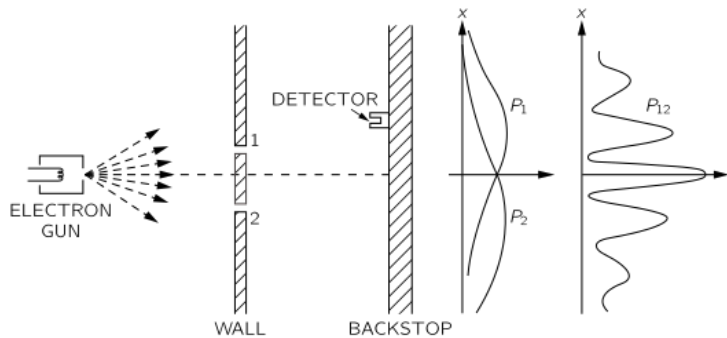


with waves (Young, 1801)

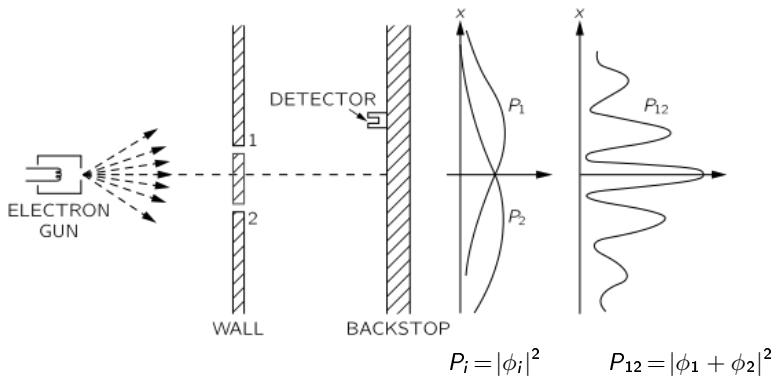


$$I_{12} = |h_1 + h_2|^2 = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

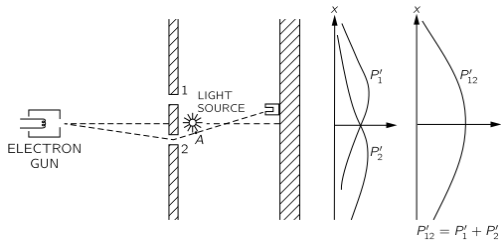
Electrons



Electrons



Electrons : Which hole? When observed -



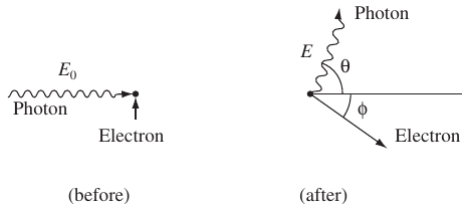
One electron double slit experiment by Akira Tonomura

<https://www.youtube.com/watch?v=jvO0P5-SMxk>

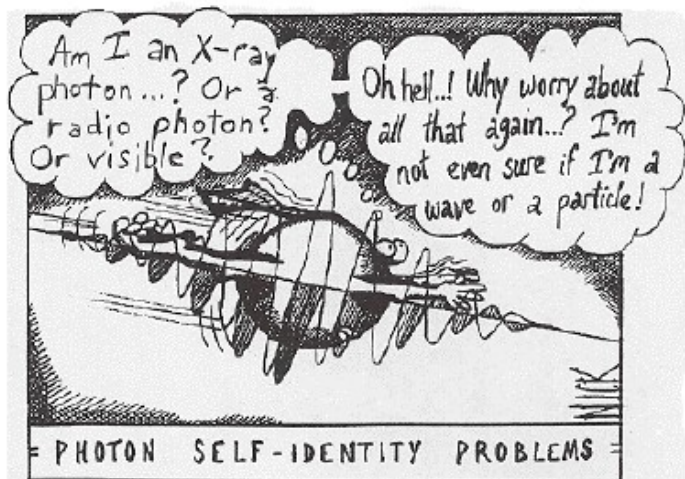
Double Slit Experiment explained! by Jim Al-Khalili

<https://www.youtube.com/watch?v=jvO0P5-SMxk>
<https://www.youtube.com/watch?v=A9tKncAdIHQ>

Compton scattering



$$\lambda - \lambda_0 = \frac{h}{m_e c} (1 - \cos \theta)$$



= PHOTON SELF-IDENTITY PROBLEMS =

